

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
Date of Assignment: 04-08-2026	Date/Time of Submission: 04-08-2026

Part B - Program

a) Create a Student Record Management PWA

Code :-

Student.cs :-

```
namespace WebApplication8.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}
```

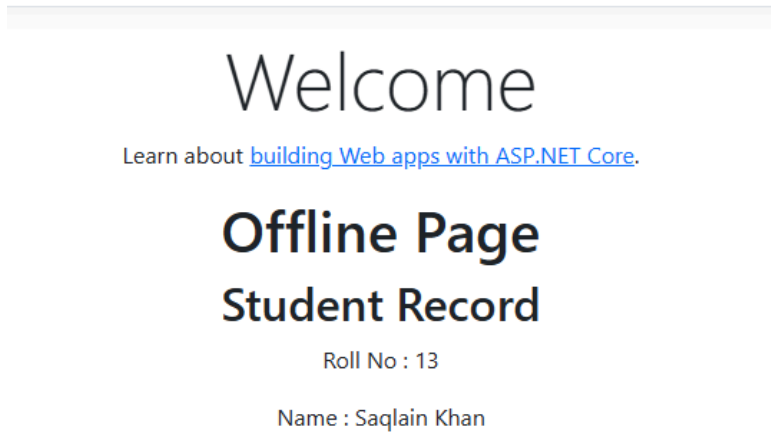
HomeController.cs :-

```
Student s = new Student();
s.RollNo = 13;
s.Name = "Saqlain Khan";
return View(s);
```

Index.cshtml :-

```
@{
    ViewData["Title"] = "Home Page";
}

<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web
apps with ASP.NET Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>
    <p>Roll No : @Model.RollNo</p>
    <p>Name : @Model.Name</p>
</div>
```

Output :-

[Prac7](#) 

- [Home](#)
- [Privacy](#)

Welcome

Learn about [building Web apps with ASP.NET Core.](#)

Offline Page

Student Record

Roll No : 13

Name : Saqlain Khan

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b) Develop a PWA that works in offline mode using Service Worker**Code :-****Offline.html :-**

```
<!DOCTYPE html>
<html>
<head>
  <title>Offline Mode</title>
  <meta charset="UTF-8">
  <link rel="manifest" href="manifest.json">
```

```
</head>
<body>
  <h1>You are currently offline</h1>
  <p>Please check your internet connection.</p>
  <script>
    if ('serviceWorker' in navigator) {
      navigator.serviceWorker.register('service-worker.js');
    }
  </script>
</body>
</html>
```

Layout.cshtml :-

```
<link rel="manifest" href="/manifest.json" />
<script>if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('/service-worker.js');
}
</script>
```

service-worker.js :-

```
const CACHE = "v1";
self.addEventListener("install", event => {
  console.log("Service Worker Installed");
  event.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/index.html"
      ]);
    })
  );
});
self.addEventListener("fetch", event => {
  event.respondWith(
    fetch(event.request)
      .catch(() => caches.match("/index.html"))
  );
});
```

manifest.json :-

```
{
  "name": "College Information Portal",
  "short_name": "PAI",
  "start_url": "/",
```

```

"display": "standalone",
"background_color": "#ffffff",
"theme_color": "#0d6efd",
"icons": [
  {
    "src": "/Icons/icon1.png",
    "sizes": "128x128",
    "type": "image/png"
  },
  {
    "src": "/Icons/icon2.png",
    "sizes": "512x512",
    "type": "image/png"
  }
]
}

```

Output :-

